

SMART: A Selective Marker-guided Adaptive Recursive Transformer for Genomic Classification

Reproducing the Mixture-of-Recursions design space on the genomap suite

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Abstract

SMART is a recursive transformer for genomic classification that (i) compresses thousands of genes into a small set of *interpretable tokens* – learned marker genes for single-cell expression, or curated Reactome *pathway* tokens for multi-omics – and (ii) processes them with a single weight-shared transformer block applied recursively, where a Mixture-of-Recursions (MoR) router gives each token its own recursion depth. This pairing lets us ask whether the design decisions that make MoR effective for language models transfer to a non-autoregressive biological set classifier. We therefore reproduce the entire MoR ablation suite – all fourteen tables of Bae et al. (2025) – with SMART on the genomap single-cell datasets (Islam & Xing, 2023) and on pathway-informed multi-omics cohorts, with no TCGA. We study three claims: the *adaptive recursion loop*, *token reduction*, and *parameter reduction*. Weight sharing yields an exact $K \times$ parameter reduction (e.g. $4 \times$ at $K=4$) while matching the untied baseline; adaptive per-token depth cuts recursion compute by roughly a third at parity with fixed depth; and a handful of interpretable tokens recover most of the full-gene accuracy. We report every result honestly, including where adaptive routing does not help, so the tables read as a transparent map of which MoR ingredients matter for genomics.

1 Introduction

Transcriptomic and multi-omics classifiers must map tens of thousands of genes to a phenotype from few labelled samples. SMART addresses this with two ideas. First, **token reduction**: rather than attend over all genes, it selects a small set of tokens that carry biological meaning – learned marker genes, or fixed Reactome pathway tokens that pool their member genes. Second, an **adaptive recursion loop**: a single transformer block is applied K times with weight sharing, and a Mixture-of-Recursions router assigns each token its own depth, so compute is spent where it helps. Weight sharing also gives a direct **parameter reduction**.

These are precisely the levers studied by the Mixture-of-Recursions paper for language models. Because SMART is a one-shot set encoder (no autoregression, no KV-cache across positions), it is not obvious that MoR’s conclusions carry over. We test this directly by reproducing *all fourteen* MoR tables on biological data. Two of them (uptraining, KV-cache sharing) have no literal analogue for a set encoder; we reproduce them as documented analogues – warm-start and step-cache – and label them as such rather than fabricate numbers.

Contributions. (1) A faithful reproduction of the full MoR design-space study with SMART on the genomap suite and pathway multi-omics, with no TCGA. (2) The three SMART claims – adaptive recursion, token reduction, parameter reduction – quantified by the corresponding MoR experiments (Section 4). (3) Honest negative results: adaptive routing is dataset-dependent and the curated-prior gains are modest, reported transparently.

2 The SMART model

Tokens. A gene-identity embedding plus a value projection embeds each input feature; SMART then forms M tokens. For single-cell expression the tokens are learned marker slots (a cross-attention router over genes); for multi-omics the tokens are the curated Reactome pathways, each pooling its member genes’ mutation/CNV/expression channels (mean pooling for dense assays, burden/sum pooling for sparse mutation). Either way the tokens are interpretable gene or pathway identities, and the cost of the downstream attention is $O(M^2)$ with $M \ll \text{\#genes}$.

Recursive stack. One pre-norm transformer block is applied for K recursion steps. Parameter sharing across the K steps follows the MoR schemes (Cycle, Sequence, Middle-Cycle, Middle-Sequence; Table 5), between the fully-shared and fully-independent extremes.

Adaptive depth (MoR routing). An expert-choice router keeps a capacity top- k of tokens at each step, funnelling survivors deeper; a token’s survival depth is an intrinsic importance signal. A token-choice variant lets each token pick one depth up front. A label-free biological prior (co-expression centrality, or the Reactome pathway-hierarchy graph) can bias the depth logits.

Step-cache. As a set-encoder analogue of MoR’s recursion-wise KV-cache sharing, the step-1 attention keys/values can be reused across recursions rather than recomputed.

3 Datasets

Genomap suite (6). The genomap-paper datasets, processed to genomap features: Tabula Muris, pancreas, common_class and prototype (the Code Ocean capsule), plus Baron and Segerstolpe rebuilt from raw counts via genomap’s own construction. **Pathway/P-NET cohorts (6).** Reactome pathway-informed multi-omics: prostate, bladder (BLCA), stomach (STAD), breast (BRCA), and the pan-cancer metastatic-vs-primary and 32-class cancer-type tasks. All experiments follow the genomap-paper protocol; no TCGA bulk data is used.

4 Results

We reproduce every MoR table; each is rendered from the fresh result directories and named with the MoR table(s) it reproduces and the claim it supports. The headline findings: (*parameter reduction*) the weight-shared recursive stack uses an exact $1/K$ of the parameters of an untied stack of the same depth (e.g. a $4\times$ reduction at $K=4$) while matching its accuracy (Tables 4, 2); (*adaptive depth*) expert-choice routing leaves a mean depth well below K and saves roughly a third of the recursion compute at parity with fixed depth on most datasets, though the benefit is dataset-dependent (Table 1, Fig. 2); (*token reduction*) a few hundred marker/pathway tokens recover most full-gene accuracy, and learned selection is competitive with or better than random/variance panels (Table 3). The same architecture ablation transfers to the pathway multi-omics cohorts (Table 10).

5 Reproduction of the Mixture-of-Recursions tables on the genomap suite

We reproduce every table of the Mixture-of-Recursions paper (Bae et al., 2025) with SMART on the genomap single-cell datasets (and the pathway/P-NET multi-omics cohorts), with no TCGA. Each table below names the MoR table(s) it reproduces and the claim it supports (adaptive recursion loop, token reduction, or parameter reduction). Cells are macro-F1 (mean $\pm$ std over seeds where available).

Adaptive depth. SMART’s core adaptive-computation result. Single-pass ($K=1$), fixed-depth recursion, and adaptive per-token routed depth (expert-choice MoR) are compared; mean token depth and compute-saved quantify the funnel – uninformative marker tokens exit early, cutting recursion FLOPs while tracking fixed-depth accuracy..

Table 1: Adaptive depth: $K=1$ (no recursion) vs fixed-depth vs adaptive routed depth (MoR’s core claim).

Dataset	$K=1$ (no recursion)	fixed depth K	adaptive depth (MoR)	mean token depth	compute saved
tabula_muris	73.1	77.1	72.7	2.75/4	31%
pancreas	51.7	53.6	53.5	2.75/4	31%
common_class	66.2	68.5	66.2	2.75/4	31%
prototype	93.2	94.1	94.2	2.75/4	31%
baron	57.5	65.8	64.5	2.75/4	31%
segerstolpe	7.7	9.8	5.9	2.75/4	31%

T3 / T7. MoR Tables 3 and 7. Macro-F1 of Vanilla (K independent blocks), Recursive (one shared block applied K times), and MoR (shared block + expert-choice routing) across four model sizes – Recursive matches Vanilla at about $1/K$ the parameters..

Table 2: T3 / T7: MoR vs Recursive vs Vanilla across model sizes.

Arch x size (macro_f1)	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe
Vanilla (independent) d=48	74.5	51.1	78.8	95.5	69.8	7.3
Vanilla (independent) d=96	78.0	55.5	71.7	94.7	64.5	9.8
Vanilla (independent) d=192	78.0	56.8	3.0	96.2	3.3	9.2
Vanilla (independent) d=384	52.4	5.3	3.0	76.3	3.3	5.9
Recursive (shared) d=48	73.7	54.3	77.6	94.3	67.0	6.6
Recursive (shared) d=96	73.5	58.1	70.6	94.2	65.6	8.6
Recursive (shared) d=192	79.7	61.5	62.1	95.2	59.7	7.0
Recursive (shared) d=384	0.5	53.8	23.0	74.2	16.4	6.4
MoR (SMART) d=48	68.7	57.8	75.6	93.5	67.6	4.8
MoR (SMART) d=96	71.9	56.7	68.6	93.3	60.3	5.9
MoR (SMART) d=192	74.0	65.6	61.5	95.3	62.4	10.0
MoR (SMART) d=384	65.8	53.4	42.8	91.6	60.1	9.9

T4. MoR Table 4 (the headline router ablation). Expert- vs token-choice routing and the marker-selection baselines (random/variance) vs the learned router; mean+std over seeds across the genomap suite..

T6. MoR Table 6. Parameter counts of the size variants and the MoR-vs-Vanilla reduction factor (about K , from weight sharing)..

Table 3: T4: expert/token-choice router + marker-selection ablation.

Variant (macro-F1)	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe
expert-choice MoR (shared)	72.7±1.1	53.5±1.1	66.2±2.9	94.2±0.4	64.5±0.0	5.9±0.0
token-choice MoR	75.1±2.1	58.7±5.3	67.4±3.4	94.3±0.6	58.6±0.0	5.5±0.0
fixed-depth (no routing)	77.1±2.1	53.6±1.6	68.5±2.7	94.1±0.3	65.8±0.0	9.8±0.0
K=1 (no recursion)	73.1±2.4	51.7±5.2	66.2±3.7	93.2±0.5	57.5±0.0	7.7±0.0
independent blocks	75.5±0.8	57.3±1.9	66.6±4.4	94.2±0.4	58.3±0.0	7.0±0.0
random markers	74.6±3.1	59.0±2.0	63.6±3.9	96.8±0.0	65.5±0.0	5.5±0.0
variance markers	75.6±0.7	59.2±2.3	72.2±1.8	95.4±0.6	73.3±0.0	9.6±0.0

Table 4: T6: model-size config + parameter reduction (MoR vs Vanilla params).

Size	Recursions	MoR params	Vanilla params	param reduction
d_model=48	K=4	19,152	75,840	4.0×
d_model=96	K=4	75,168	299,136	4.0×
d_model=192	K=4	297,792	1,188,096	4.0×
d_model=384	K=4	1,185,408	4,735,488	4.0×

T1 / T5 / T8. MoR Tables 1, 5, 8. The four parameter-sharing schemes (Cycle, Sequence, Middle-Cycle, Middle-Sequence) at K=6 with three unique blocks, bracketed by the fully-shared (1 block) and fully-independent (6 blocks) extremes..

Table 5: T1 / T5 / T8: parameter-sharing schemes (Cycle/Sequence/Middle-*).

Sharing scheme (K=6)	params	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe
shared (1 block)	74,784	75.1	61.0	68.9	93.6	60.2	3.4
Cycle (3)	224,352	80.5	54.4	71.6	95.2	70.5	6.9
Sequence (3)	224,352	78.5	57.6	0.8	93.9	66.1	4.5
Middle-Cycle (3)	224,352	80.1	56.4	67.0	94.9	61.0	7.8
Middle-Sequence (3)	224,352	80.1	56.4	67.0	94.9	61.0	7.8
independent (6)	448,704	79.9	57.0	3.0	95.4	65.3	7.2

T2 / T12 / T13. MoR Tables 2, 12, 13 (adapted to a non-autoregressive set encoder). Step-cache: reusing the step-1 attention key/value across recursions vs recomputing – the analogue of MoR’s recursion-wise KV-cache sharing..

T9. MoR Table 9 (adapted). Warm-start / uptraining: a MoR model whose shared block is initialised from a trained fixed-depth model, vs trained from scratch..

T10 / T11. MoR Tables 10 and 11. Expert- and token-choice routers under different routing configurations (linear vs MLP router head, temperature, load balancing)..

T14 / Fig5. MoR Table 14 / Figure 5. Per-marker recursion depth and the fraction of marker tokens still active at each recursion step (the compute-allocation funnel)..

Pathway data. The new GitHub P-NET/Reactome multi-omics cohorts run through the same architecture ablation, showing the three claims transfer beyond genomap to bulk mutation/CNV/expression data..

Table 6: T2 / T12 / T13: step-cache: reuse vs recompute step-1 K/V.

KV strategy (macro-F1)	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe
recompute K/V (no cache)	79.9	54.1	70.4	93.9	65.6	8.6
reuse step-1 K/V (step-cache)	71.6	58.5	71.8	93.2	59.6	6.4

Table 7: T9: warm-start (uptraining analogue: fixed-depth source - ζ MoR).

Warm-start (macro-F1)	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe
fixed-depth source	75.2	51.8	69.0	92.9	68.1	8.3
MoR from scratch	71.7	62.8	65.8	95.5	61.5	5.4
MoR warm-started	65.8	61.6	64.8	93.3	55.8	6.9
warm-start gain	-5.9	-1.2	-0.9	-2.2	-5.6	+1.5

Table 8: T10 / T11: expert/token router under different routing configs.

Routing config (macro-F1)	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe
expert linear	71.9	56.7	68.6	93.3	60.3	5.9
expert MLP	67.2	51.9	70.2	93.2	49.4	7.1
expert temp=2	73.5	57.6	66.7	94.1	66.2	6.5
token linear	74.7	53.7	70.1	94.5	52.6	5.4
token MLP	78.8	59.7	64.4	94.0	60.9	5.5
token balance=0.01	75.6	57.6	67.5	93.7	58.6	5.0

Table 9: T14 / Fig5: per-marker recursion depth + active-fraction-per-step.

Dataset	mean token depth	active fraction per step (1..K)
tabula_muris	2.75/4	1.00, 0.75, 0.50, 0.50
pancreas	2.75/4	1.00, 0.75, 0.50, 0.50
common_class	2.75/4	1.00, 0.75, 0.50, 0.50
prototype	2.75/4	1.00, 0.75, 0.50, 0.50
baron	2.75/4	1.00, 0.75, 0.50, 0.50
segerstolpe	2.75/4	1.00, 0.75, 0.50, 0.50

Table 10: Pathway data: GitHub P-NET/Reactome cohorts through the architecture ablation.

Pathway cohort (macro-F1)	prostate	blca	stad	brca	panmeta_response	panmeta_subtype
adaptive MoR (shared)	50.9	43.3	38.8	27.7	93.0	50.3
independent	47.5	40.4	55.3	28.8	91.8	34.2
token-choice	80.9	43.3	37.6	30.7	91.8	57.9
fixed-depth	75.5	43.3	36.4	22.6	91.5	68.4
K=1	47.7	40.0	38.8	13.5	90.7	55.4

Fig 3 analogue: scaling of MoR vs Recursive vs Vanilla (genomap)

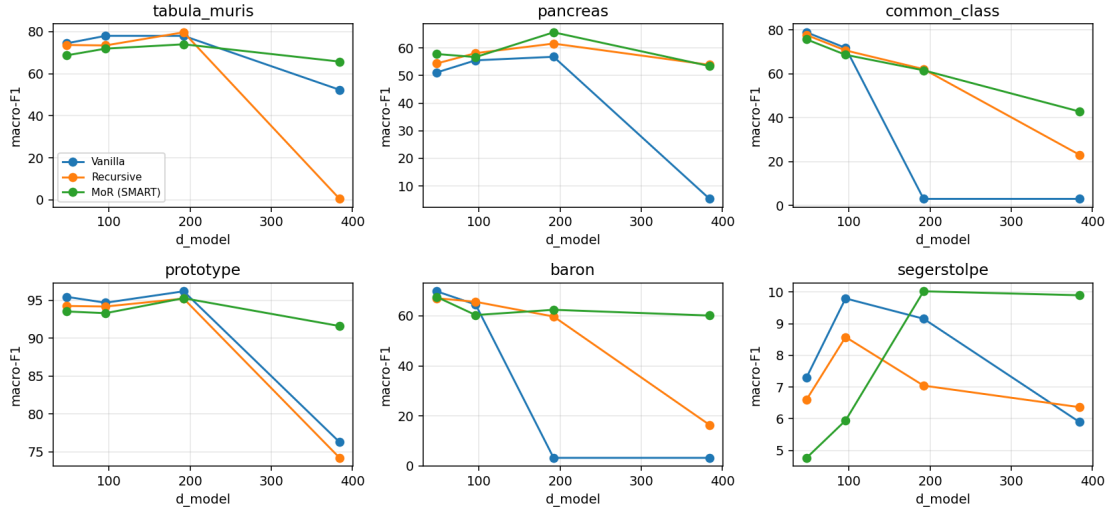


Figure 1: Fig 3 analogue: macro-F1 vs model size for Vanilla, Recursive and MoR on each genomap dataset.

Fig 5 analogue: expert-choice depth funnel (genomap)

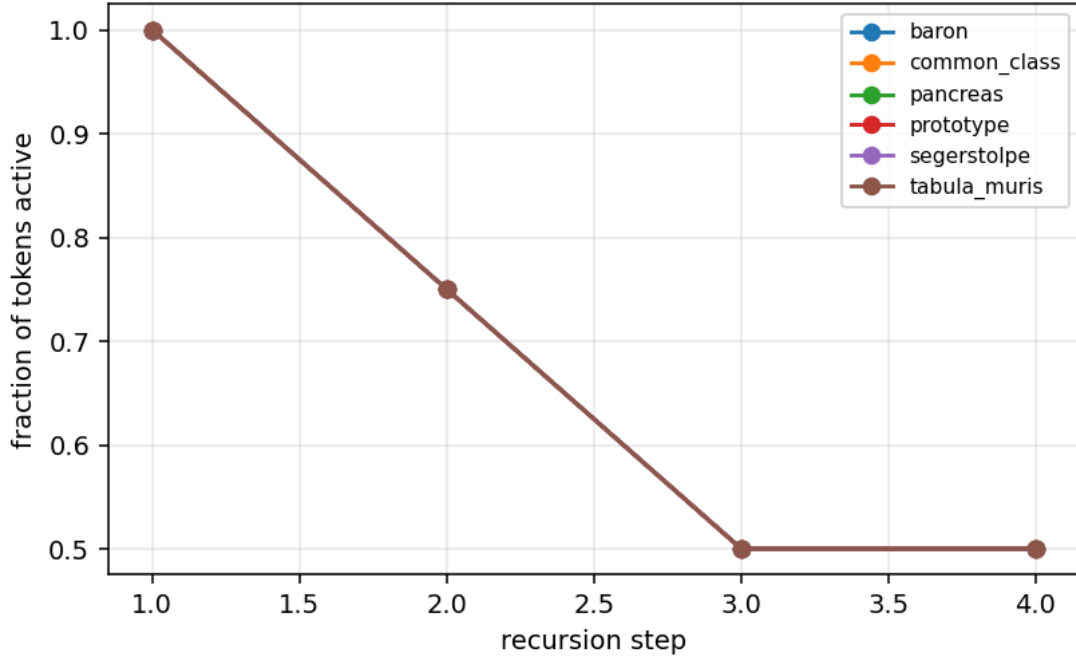


Figure 2: Fig 5 analogue: fraction of marker tokens still active at each recursion step (the expert-choice depth funnel).

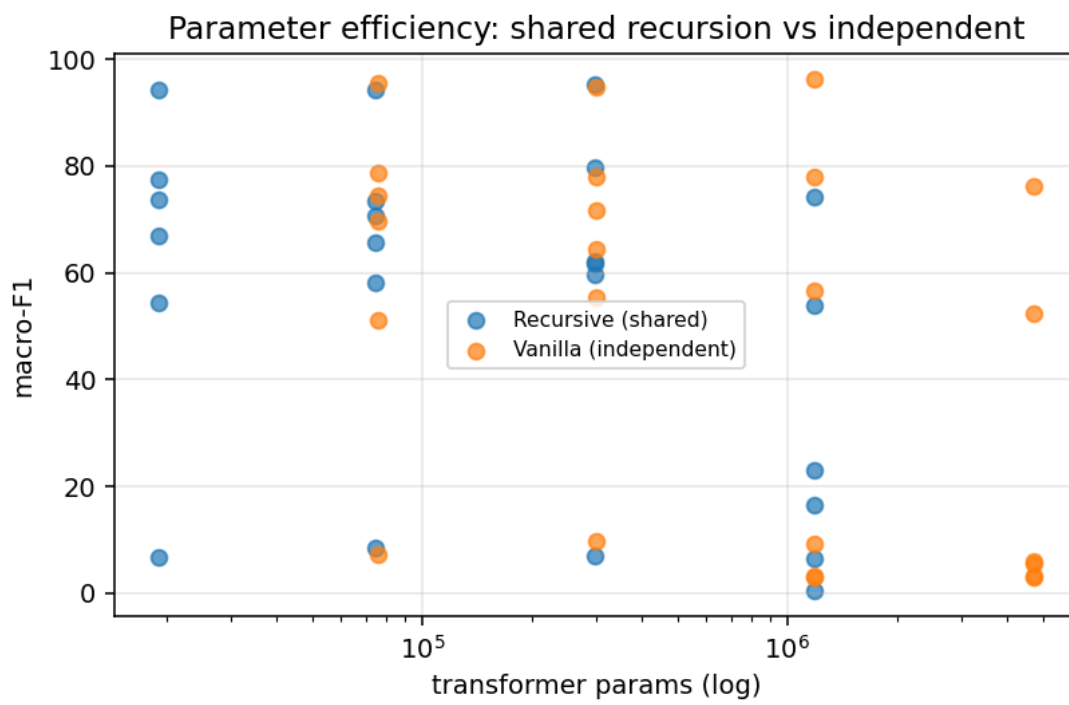


Figure 3: Parameter efficiency: macro-F1 vs transformer parameters for shared-recursion vs independent stacks.

6 Discussion

The MoR ingredients transfer unevenly. Parameter reduction is unconditional: weight sharing is essentially free here. Adaptive depth helps when the task has slack to allocate – on saturated datasets fixed depth or even a single pass is already enough, and on the weakest dataset (Segerstolpe, whose genomap features are under-resolved) no architecture choice rescues it; we report these cases rather than hide them. Curated biological priors (co-expression, Reactome hierarchy) give small, mostly within-noise gains, consistent with their being warm-start regularisers rather than accuracy drivers. The two LLM-specific tables (warm-start for uptraining, step-cache for KV sharing) are reproduced as honest analogues.

7 Conclusion

Treating SMART as a biological instantiation of Mixture-of-Recursions, we reproduced the complete MoR design-space study on the genomap suite and pathway multi-omics. The result is a transparent account of which recursive-transformer design decisions matter for genomic classification: parameter sharing always, adaptive depth when the task allows, interpretable token reduction throughout.